

LAB NO.: 11
DISK SCHEDULING ALGORITHM

Lab Exercises:

1. Write a C/C++ Program to simulate the following algorithms find the total no. of cylinder movements for various input requests

(i) FCFS (ii) SSTF (iii) SCAN (iv) C-SCAN

Program:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
void sort(int arr[], int n) {
    int i, j, temp;
    for(i = 0; i < n - 1; i++) {
        for(j = 0; j < n - i - 1; j++) {
            if(arr[j] > arr[j + 1]) {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}
```

```
void fcfs(int req[], int n, int head) {
    int i, total = 0;
    for(i = 0; i < n; i++) {
        total += abs(req[i] - head);
        head = req[i];
    }
    printf("FCFS Total Head Movement = %d\n", total);
}
```

```
void sstf(int req[], int n, int head) {
    int visited[50] = {0};
    int i, count = 0, total = 0;

    while(count < n) {
        int min = 10000, index = -1;
        for(i = 0; i < n; i++) {
            if(!visited[i]) {
                int dist = abs(req[i] - head);
                if(dist < min) {
                    min = dist;
                    index = i;
                }
            }
        }
        visited[index] = 1;
```

```

        total += min;
        head = req[index];
        count++;
    }
    printf("SSTF Total Head Movement = %d\n", total);
}

```

```

void scan(int req[], int n, int head, int disk_size) {
    int i, total = 0;
    int arr[50];

    for(i = 0; i < n; i++)
        arr[i] = req[i];

    arr[n] = disk_size - 1;
    n++;

    sort(arr, n);

    int pos;
    for(i = 0; i < n; i++) {
        if(head < arr[i]) {
            pos = i;
            break;
        }
    }

    for(i = pos; i < n; i++) {
        total += abs(arr[i] - head);
        head = arr[i];
    }

    for(i = pos - 1; i >= 0; i--) {
        total += abs(arr[i] - head);
        head = arr[i];
    }

    printf("SCAN Total Head Movement = %d\n", total);
}

```

```

void cscan(int req[], int n, int head, int disk_size) {
    int i, total = 0;
    int arr[50];

    for(i = 0; i < n; i++)
        arr[i] = req[i];

    arr[n] = disk_size - 1;
    arr[n + 1] = 0;
    n += 2;

    sort(arr, n);

```

```

int pos;
for(i = 0; i < n; i++) {
    if(head < arr[i]) {
        pos = i;
        break;
    }
}

for(i = pos; i < n; i++) {
    total += abs(arr[i] - head);
    head = arr[i];
}

for(i = 0; i < pos; i++) {
    total += abs(arr[i] - head);
    head = arr[i];
}

printf("C-SCAN Total Head Movement = %d\n", total);
}

int main() {
    int n, i, head, disk_size;

    printf("Enter number of requests: ");
    scanf("%d", &n);

    int req[50];

    printf("Enter request sequence:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &req[i]);

    printf("Enter initial head position: ");
    scanf("%d", &head);

    printf("Enter disk size: ");
    scanf("%d", &disk_size);

    fcfs(req, n, head);
    sstf(req, n, head);
    scan(req, n, head, disk_size);
    cscan(req, n, head, disk_size);

    return 0;
}

```

Output:

```
• (base) mca@computinglab25-39:~/Param_240905320/Lab11$ gcc Lab11_1.c -o Lab11_1
• (base) mca@computinglab25-39:~/Param_240905320/Lab11$ ./Lab11_1
Enter number of requests: 8
Enter request sequence:
98 183 37 122 14 124 65 67
Enter initial head position: 53
Enter disk size: 200
FCFS THM = 640
SSTF THM= 236
SCAN THM = 331
C-SCAN THM = 382
```

2. Write a C/C++ Program to simulate the following algorithms find the total no. of cylinder movements for various input requests

(i) LOOK (ii) C-LOOK

Program:

```
#include <stdio.h>
#include <stdlib.h>

void sort(int arr[], int n)
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            if (arr[i] < arr[j])
            {
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
    }
}

void clook(int req[], int n, int head, int prev_head, int disk_size)
{
    int arr[50];

    for (int i = 0; i < n; i++)
        arr[i] = req[i];

    sort(arr, n);

    int direction = (prev_head > head) ? 0 : 1;
    int total = 0, index, i;

    for (i = 0; i < n; i++)
    {
        if (arr[i] >= head)
        {
            index = i;
            break;
        }
    }

    if (i == n)
        index = n;

    if (direction == 1)
    {
        for (i = index; i < n; i++)
```

```

    {
        total += abs(head - arr[i]);
        head = arr[i];
    }

    if (index > 0)
    {
        total += abs(head - arr[0]);
        head = arr[0];
    }

    for (i = 1; i < index; i++)
    {
        total += abs(head - arr[i]);
        head = arr[i];
    }
}
else
{
    for (i = index - 1; i >= 0; i--)
    {
        total += abs(head - arr[i]);
        head = arr[i];
    }

    if (index < n)
    {
        total += abs(head - arr[n - 1]);
        head = arr[n - 1];
    }

    for (i = n - 2; i >= index; i--)
    {
        total += abs(head - arr[i]);
        head = arr[i];
    }
}

printf("\nC-LOOK THM: %d \n", total);
}

void look(int req[], int n, int head, int prev_head, int disk_size)
{
    int arr[50];
    for (int i = 0; i < n; i++)
    {
        arr[i] = req[i];
    }
    sort(arr, n);
    int direction = (prev_head > head) ? 0 : 1;
    int i, total = 0, index;
    for (i = 0; i < n; i++)

```

```

{
    if (arr[i] >= head)
    {
        index = i;
        break;
    }
}
if (i == n)
{
    index = n;
}
if (direction == 1)
{
    for (i = index; i < n; i++)
    {
        total += abs(head - arr[i]);
        head = arr[i];
    }
    for (i = index - 1; i >= 0; i--)
    {
        total += abs(head - arr[i]);
        head = arr[i];
    }
}
else
{
    for (i = index - 1; i >= 0; i--)
    {
        total += abs(head - arr[i]);
        head = arr[i];
    }
    for (i = index; i < n; i++)
    {
        total += abs(head - arr[i]);
        head = arr[i];
    }
}
printf("Look THM: %d", total);
}

```

```

int main()
{
    int n, i, head, prev_head, disk_size;

    printf("Enter number of requests: ");
    scanf("%d", &n);

    int req[50];

    printf("Enter request sequence:\n");
    for (i = 0; i < n; i++)
        scanf("%d", &req[i]);
}

```

```
printf("Enter initial head position: ");
scanf("%d", &head);
printf("Enter previous head position: ");
scanf("%d", &prev_head);

printf("Enter disk size: ");
scanf("%d", &disk_size);

look(req, n, head, prev_head, disk_size);
clock(req, n, head, prev_head, disk_size);

return 0;
}
```

Output:

```
Enter number of requests: 8
Enter request sequence:
98
183
37
122
14
124
65
67
Enter initial head position: 53
Enter previous head position: 125
Enter disk size: 200
Look THM: 208
C-LOOK THM: 326
```